5 **(a)** A thermometer and an electrical heater are inserted into holes in an aluminium block of mass 960 g, as shown in Fig. 5.1.

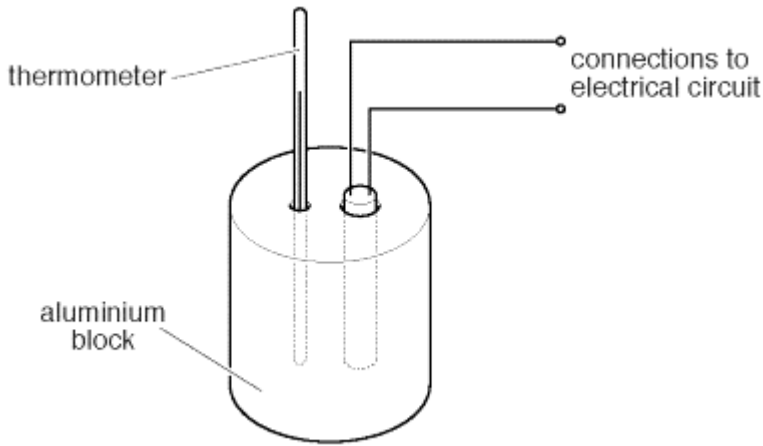

Fig. 5.1

The power rating of the heater is 54 W. The heater is switched on and readings of the temperature of the block are taken at regular time intervals. The variation with time t of the temperature θ of the block is shown in Fig. 5.2.

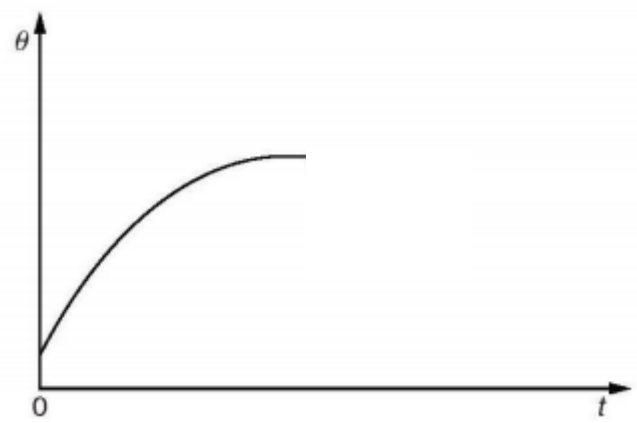

Fig. 5.2

(i) Suggest why the rate of rise of temperature of the block decreases to zero.

.....

.....

..... [2]

(ii) When the block reaches a constant temperature, the heater is switched off and then further temperature readings are taken. The maximum rate of fall of temperature is found to be around 3.7 °C per minute.

Estimate the heat capacity of the aluminium block.

heat capacity = …….J K^{-1} [2]

(b) Fig. 5.3 shows an insulated cylinder fitted with a perfectly fitting piston and a heater. There is negligible friction between the cylinder and the piston. The cylinder contains a fixed mass of an ideal gas.

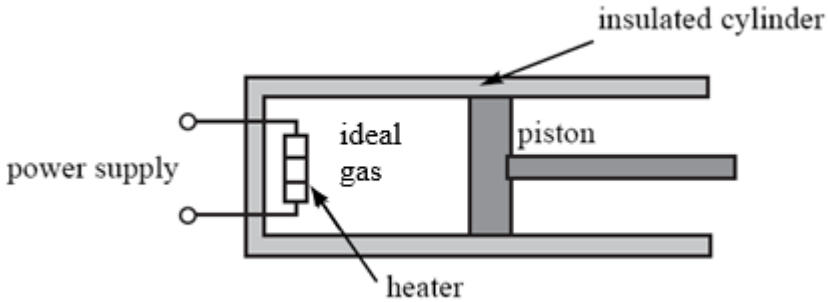

Fig. 5.3

The following two experiments are performed.

Experiment 1 The heater provides 150 J of energy with the piston fixed in position. The temperature rise of the gas is found to be 29 K.

Experiment 2 The heater supplies 150 J of energy with the piston free to move so that the gas expands at constant pressure. The temperature rise of the gas is now 18 K.

(i) Explain why the temperature rise is different in the two experiments.

.....

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..... [2]

(ii) Fill up the table below to show the values of each of the quantities for experiments 1 and 2. Present your workings using the space provided below.

	thermal energy supplied to gas / J	work done on gas / J	increase in internal energy of gas / J
Experiment 1			
Experiment 2			

[4]